

Heterogeneous Catalysis by Metals

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1 INTRODUCTION

Catalysis refers to the phenomenon by which the rate of a chemical reaction is accelerated by a substance (the catalyst) not appreciably consumed in the process. The term 'catalysis' was coined by Berzelius in 1835 and scientifically defined by Ostwald in 1895, but applications based on catalysis can be traced back to thousands of years ago with the discovery of fermentation to produce wine and beer.¹ Nowadays, catalysts are used in 80% of all chemical industrial processes, and create annual global sales of about 1500 billion dollars and contribute directly or indirectly to approximately 35% of the world's GDP.² Catalysis is central to a myriad of applications, including the manufacture of commodity, fine, specialty, petro-, and agro- chemicals as well as the production of pharmaceuticals, cosmetics, foods, and polymers. Catalysis is also an important component in new processes for the generation of clean energy, and in the protection of the environment both by abating environmental pollutants and by providing alternative cleaner chemical synthetic procedures.

Most catalytic processes are heterogeneous in nature, typically involving a solid catalyst and gas or liquid phase reactants. When compared with homogeneous catalysts, heterogeneous catalysts offer inherent advantages because of their ease of preparation, handling, separation from the reaction mixture, recovery, and reuse, and also in terms of their stability, low cost, and low toxicity. Heterogeneous catalysts are most commonly made out of metals and metal oxides, although metal sulfides, nitrides, carbides, phosphates and phosphides, ion-exchange resins, and clays are also employed in selected applications. Metal catalysts, especially those containing transition metals, have proven particularly useful in industrial catalysis, because they have the ability to easily activate key molecules such as H₂, O₂, N₂, and CO as well as polyatomic organic molecules with C–H, C–O, C–N, and

C–Cl bonds.³ Table 1 summarizes some of the most important metal-based catalytic processes developed since the 1870s.^{1,4}

In the following sections, the basic chemical concepts behind heterogeneous metal catalysis will be introduced. Then, typical examples of practical catalytic systems related to the generation of energy, the production of chemicals, the preparation of materials, and the cleaning of the environment will be provided. Finally, several promising directions related to heterogeneous metal catalysis will be highlighted.

2 FUNDAMENTALS OF HETEROGENEOUS METAL CATALYSIS

2.1 Kinetics Versus Thermodynamics, Active Centers, Catalytic Cycles

Catalysis relies on changes in the kinetics of chemical reactions. Thermodynamics acts as an arrow to show the way to the most stable products, but kinetics defines the relative rates of the many competitive pathways available for the reactants, and can therefore be used to make metastable products from catalytic processes in a fast and selective way. Indeed, catalysts work by opening alternative mechanistic routes with lower activation energy barriers than those of the noncatalyzed reactions.⁵ As an example, Figure 1 illustrates how the use of metal catalysts facilitates the dissociation of molecular oxygen, and with that the oxidation of carbon monoxide. Thanks to the availability of new pathways, catalyzed reactions can be carried out at much faster rates and at lower temperatures than noncatalyzed reactions. Note, however, that a catalyst can shorten the time needed to achieve thermodynamic equilibrium, but cannot shift the position of that equilibrium, and therefore cannot catalyze a thermodynamically unfavorable reaction.⁶

The new mechanistic routes opened by the introduction of a catalyst to the reaction mixture typically start with the adsorption of the reactants. Because of the heterogeneous nature of the surface of the catalyst, both the adsorption and the subsequent conversion reactions may take place preferentially at particular ensembles of surface sites, often called active sites or active centers.^{7,8} An atomic ensemble may become active because of a specific structural arrangement of those atoms. Alternatively, the electronic properties of metal atoms may also influence the adsorption and activation of reactants. Typically, the performance of catalytic active sites depends on both structural and electronic effects.

After diffusion towards and adsorption on the surface-active sites, the reactants are converted to products. Those products then desorb and diffuse out of the catalyst, leaving the active centers available for new incoming reactants.⁷ That way, the catalytic cycle can be repeated many times on each active site. For this to work, however, the bond strength between the adsorbed surface species and the active sites

Table 1 Important industrial heterogeneous catalytic processes promoted by metals^{1,4}

Process	Year	Main uses	Key reaction scheme	Typical catalyst
SO ₂ oxidation to sulfuric acid	1875	Chemicals, Metallurgic processing	$\text{SO}_2 + 1/2\text{O}_2 \rightarrow \text{SO}_3$	Pt on asbestos, MgSO ₄ , or SiO ₂ (replaced by V ₂ O ₅ – K ₂ SO ₄ /SiO ₂ since the 1920s)
Methanol to formaldehyde	1890	Resins for adhesives	$\text{CH}_3\text{OH} + 1/2\text{O}_2 \rightarrow \text{HCHO} + \text{H}_2\text{O}$	Ag wire gauze or Ag crystals
Olefin hydrogenation	1902	Oil refining	$\text{C}_2\text{H}_4 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$	Ni, Pt
Hydrogenation of edible fats and oils	1900s	Food production	unsaturated fatty acids $\rightarrow$ partially saturated acids	Ni on a support
Methanation	1900s	Fuels	$\text{CO} + 3\text{H}_2 \rightarrow \text{CH}_4 + \text{H}_2\text{O}$	Ni on Al ₂ O ₃ or other oxide supports
Ammonia synthesis (Haber)	1913	Fertilizers	$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$	Fe promoted with Al ₂ O ₃ , K ₂ O, CaO, and MgO
Ammonia oxidation (Ostwald)	1906	Nitric acid production	$4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$	90%Pt-10% Rh wire gauze
Fisher–Tropsch synthesis	1938	Fuels	$\text{CO} + \text{H}_2 \rightarrow \text{paraffins}$	Fe or Co with promoters on support
Steam reforming	1926	Synthesis gas	$\text{C}_n\text{H}_m + n\text{H}_2\text{O} \rightarrow n\text{CO} + [n + (m/2)]\text{H}_2$	Ni on support promoted by K ₂ O
Ethylene to ethylene oxide	1937	Antifreeze	$\text{C}_2\text{H}_4 + 1/2\text{O}_2 \rightarrow (\text{CH}_2)_2\text{O}$	Ag on α -Al ₂ O ₃ , promoted by Cl and Cs
Hydrogen cyanide synthesis	1930s	Chemicals	$\text{CH}_4 + \text{NH}_3 + 3/2\text{O}_2 \rightarrow \text{HCN} + 3\text{H}_2\text{O}$	90%Pt-10% Rh wire gauze
Catalytic reforming	1940s	Fuels	e.g., $n\text{-C}_6\text{H}_{14} \rightarrow i\text{-C}_6\text{H}_{14}$	Pt, Pt–Re or Pt–Sn on acidified Al ₂ O ₃ or zeolite
Benzene to cyclohexane	1940s	Nylon	$\text{C}_6\text{H}_6 + 3\text{H}_2 \rightarrow \text{C}_6\text{H}_{12}$	Ni, Pt, or Pd catalysts
Vinyl acetate synthesis	1968	Polymers	$\text{C}_2\text{H}_4 + \text{CH}_3\text{COOH} + 1/2\text{O}_2 \rightarrow \text{CH}_3\text{COOCH}=\text{CH}_2 + \text{H}_2\text{O}$	Pd on SiO ₂ or α -Al ₂ O ₃
Automobile three-way catalysis	1970s	Pollution control	$\text{CO} + \text{H.C.} + \text{NO}_x + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{N}_2$	Pt, Pd, Rh on monolith support

needs to be neither too weak nor too strong: too weak and the reactants cannot be readily activated; too strong and either the reactants are completely decomposed on the surface or the products cannot desorb.⁶ This relationship between adsorption energy and catalytic performance is epitomized by volcano plots such as those in Figure 2.^{9–12} In many instances, the adsorption energy can also be correlated to specific properties of the metals such as their d-band occupancy (%d) character (Figure 2b).¹²

2.2 Activity, Selectivity, Stability

Three key parameters, activity, selectivity, and stability, can be used to evaluate catalyst performance. Achieving good and sustainable activity and selectivity at low cost is an everlasting goal in catalyst design and development. Activity refers to the rate at which a given reaction proceeds. In practice, activity can be expressed in absolute terms, but is often reported using relative units such as specific rate (rate divided by catalyst weight), areal rate (rate divided by surface area), turnover frequency (molecules converted per active center per unit

time), conversion after specified time (fraction of reactants converted per fixed time), or required temperature for a given conversion under specific conditions.¹³ In industry, activity is often expressed by space-time yield, the amount of product formed per unit time per unit volume of reactor. Regardless of the parameters adopted to express activity, however, it needs to be appreciated that such activity depends on the catalyst load in the reactor, the concentration of the reactants, and the temperature, pressure, and flow during operation.

In addition to activity, the design of catalytic processes needs to consider their selectivity towards the desirable products.¹⁴ In fact, selectivity is arguably the most important criterion to take into account to decide on a particular catalytic process. Selective reactions consume less reactant, avoid the need of costly separations, and do not produce potentially polluting by-products. Unfortunately, it is not easy to design selective catalytic processes from first principles. For one, a catalyst active for a given reaction may be inactive for a closely related one. In addition, given a set of reactants, the use of different catalysts may lead to different products. For example, ethanol is dehydrogenated to acetaldehyde on Cu

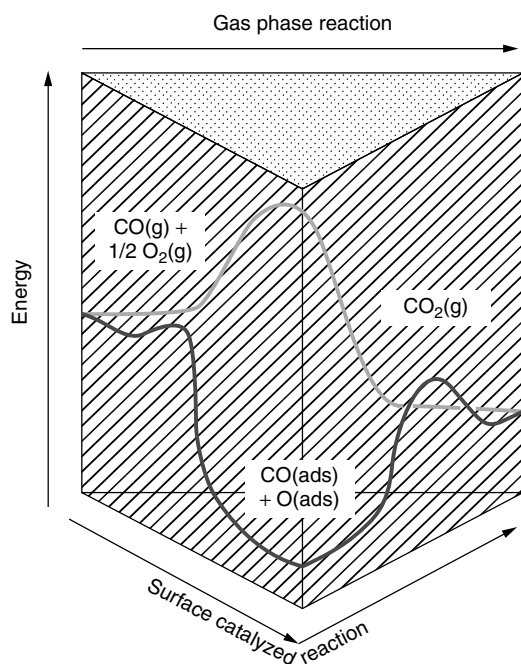


Figure 1 Energy diagram illustrating how solid catalysts facilitate reactions by opening new mechanistic routes. This example corresponds to the oxidation of carbon monoxide, a conversion that can be accelerated by previous dissociation of the molecular oxygen on many transition metal surfaces

but dehydrated to ethylene or ethyl ether on γ - Al_2O_3 .⁴ Many competing reactions may also be promoted by one particular

catalyst, in which case the distribution of the resulting products may depend on experimental conditions such as reaction temperature, pressure, and contact time.

Finally, there is the issue of stability. Ideally, catalysts are expected not to be consumed and to maintain a constant level of performance during the course of the reaction. In reality, however, both catalytic activity and catalytic selectivity may deteriorate during operation. The lifetime of most industrial catalysts is finite, ranging from months to years. The most common reasons for this catalyst deactivation include the evaporation, washout, reduction, corrosion or other transformation of the active catalytic species during reaction, the formation of carbonaceous deposits (coking), the poisoning due to strong adsorption of impurities such as sulfur or carbon monoxide on the surface of the catalyst, and/or the coalescence of metal particles (sintering).^{4,13} Some of these processes are reversible, in which case the catalysts can be regenerated by special treatments in situ in the reactor such as burning off of the coke deposits or other in situ chemical treatments, but many are irreversible and require the replacement of the catalyst. Fluctuations in processing conditions may influence the stability of catalysts as well, so engineering optimization is also crucial in the design of catalytic processes.

2.3 Supports, Promoters

As mentioned above, heterogeneous catalysis is centered around the chemistry of adsorbed species on solid surfaces. Consequently, it is important to maximize the surface-to-volume ratio of the active phase to minimize cost and

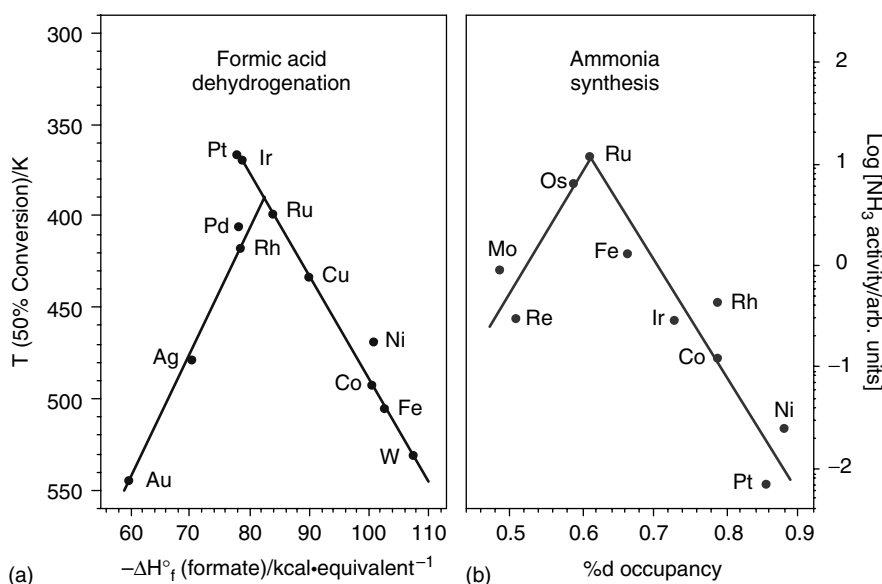


Figure 2 Volcano plots illustrating how catalytic activities depend on the nature of the metal used as catalyst. (a) Correlation between catalytic activity for formic acid dehydrogenation and enthalpy of formation of metal formates. (Ref. 10. Reproduced by permission of John Wiley & Sons, Inc.)^{9,10} (b) Correlation between catalytic activity for ammonia synthesis and degree of d-band filling in the metal used as catalyst. (Ref. 12. Reproduced by permission of John Wiley & Sons, Inc.)¹¹

improve performance. Because of this, the direct use of metal wires, gauzes, and foils in industrial catalysis is rare. The surface area of bulk metals is rather small, with only a few active sites exposed, and pure metals in high surface area forms are generally not thermally stable under catalytic conditions. Instead, metals are typically dispersed onto high surface area refractory supports such as alumina, silica, zeolites, activated carbon, titania, zirconia, or mixed oxides. Pillared clays, mesoporous materials, ceramics, silicon carbide, graphite, carbon nanofibers, magnesia, metal fluorides, calcium carbonate, and barium sulfate have also been used occasionally for this purpose. Figure 3 shows a transmission electron microscopy (TEM) picture of a typical catalyst; in this case silver particles approximately 10–20 nm in diameter are deposited on a porous alumina support.¹⁵ The choice of support is primarily based on its surface area, but thermal and chemical stability, chemical properties, mechanical strength, and price also need to be considered.¹³ Although most supports are used primarily to disperse a catalytic active phase, they often add to the overall chemistry of the catalytic process. In particular, supports with different acid-base or redox properties often exert marked influence on catalytic behavior. Zeolites (*see Zeolites*) can also provide unique shape selectivity, since the uniformity of their pores can regulate the relative flows of the entering reactants, the leaving products, and the transition states of the reaction based on their sizes, thus determining the product distribution.¹⁶ Last but not least, strong interactions between metal catalysts and supports such as titania (SMSI) can significantly modify the electronic and chemical properties of the former.⁴

To further optimize the performance of metal catalysts, promoters are often added. Textural promoters work by separating metal particles from one another to minimize sintering, whereas electronic and structural promoters change the electronic or crystal structure of the active metal. For instance, a small amount of potassium acetate is added to

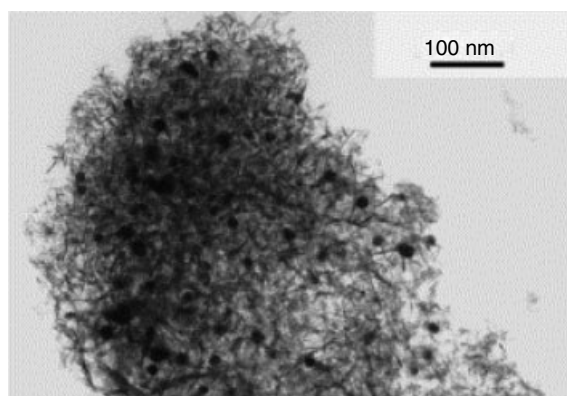


Figure 3 TEM picture of a typical supported metal catalyst, silver nanoparticles (dark spots) deposited on a high surface area alumina support. (Reprinted from Ref. 15. © 1998, with permission from Elsevier)

the Pd/SiO₂ or Pd/ α -Al₂O₃ catalysts used in the synthesis of vinyl acetate to promote the adsorption of acetic acid and lower the barrier to vinyl acetate formation, thus increasing the overall activity and minimizing the yield of CO₂.¹⁷ The addition of potassium compounds or other alkaline substances to nickel catalysts during steam reforming can facilitate both the adsorption of water and the removal of carbonaceous deposit.⁴ In ammonia synthesis, Al₂O₃, CaO, and MgO are added to the iron catalyst as textural promoters to minimize the sintering of Fe, and K₂O as an electronic promoter to increase the activity for the dissociative adsorption of N₂.⁴ Nowadays, almost all catalytic processes in industry involve the incorporation of promoters during catalyst preparation and/or the feed of additives with the reaction mixture.

2.4 Preparation and Characterization of Metal Catalysts

Because of the complexity of solid catalysts, where the active phase is dispersed onto a high surface area solid and promoted with a number of additives, the preparation of catalysts is still somewhat of an art. Numerous methods have been developed over the years for the preparation of metal catalysts. For example, some iron-based catalysts for ammonia synthesis are manufactured by fusion of Fe₃O₄ with small amounts of K₂CO₃, Al₂O₃, and other ingredients.⁴ The porous Raney nickel catalysts used for the selective hydrogenation of organic chemicals are prepared by alloying Ni with Al and then leaching away the Al in a boiling NaOH solution.¹⁷ The silver catalyst used in the partial oxidation of methanol is made electrochemically using an AgNO₃ electrolyte.¹⁷ However, the most common way to prepare supported metal catalysts is by impregnation, where a porous support is soaked into a solution containing a metal precursor such as HAuCl₄ or H₂PtCl₆.¹⁸ Coprecipitation is another traditional method where a solution containing a soluble metal salt and a precursor for the support such as Al(NO₃)₃ or ZrOCl₂ are thoroughly mixed with an aqueous inorganic base to precipitate the cations as hydroxides or carbonates.¹⁹ Sol-gel methods (*see Sol-Gel Synthesis of Solids*) involving the hydrolysis of metal alkoxides such as Si(OC₂H₅)₄ and Al(i-OC₃H₇)₃ can also be used to form high surface area supports.²⁰ In all of these three cases, the resulting materials need to be calcined and reduced in order to expose the metallic phase. Decarbonylation of metal carbonyls adsorbed on appropriate supports, oxidative transformation of amorphous alloys into supported metal catalysts, and flame spray pyrolysis, are also viable routes to catalyst preparation.¹⁷ Finally, for industrial applications, the powder catalysts obtained are further shaped into tablets, pellets, rings, extrudates, spheres, shells, or granules (Figure 4).

The processes used for the preparation of heterogeneous catalysts are not only versatile, but also complex; even subtle changes in preparation conditions often lead to clear differences in catalyst performance. Therefore, it is highly desirable to develop a microscopic understanding of the

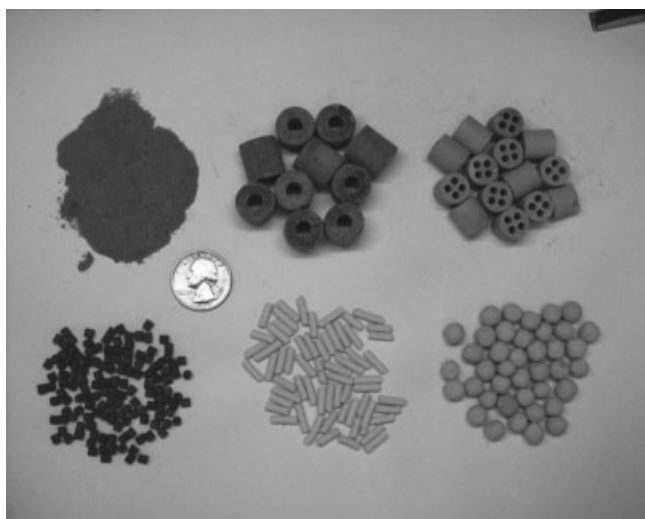


Figure 4 Examples of the final forms of the catalysts used in industrial processes. Shown are powders, rings, four-hole cylinders, tablets, pellets, and spheres of different supported metal catalysts

relation between structure and reactivity in catalysts. For now, however, the design of catalyst preparation from first principles is still not possible. Luckily, a number of analytical methods are available for the characterization of the final solids.²¹ X-ray diffraction (XRD) provides information on the bulk structure, degree of crystallinity, and average crystal size of supported catalysts. Electron microscopy, in either transmission (TEM) or scanning (SEM) modes, aids in the determination of individual metal particle sizes and morphology, and of metal dispersion. Nitrogen adsorption isotherms are used to derive surface areas, pore size distributions, and pore volumes. X-ray photoelectron spectroscopy (XPS) facilitates the identification of surface species and their oxidation states. Infrared spectroscopy of probe adsorbates such as CO, NO, NH₃, C₂H₄, CH₃OH, and pyridine gives information on the nature of adsorption sites and reaction mechanism on catalyst surfaces.²² Finally, chemical titration can be used to identify specific properties of active sites. For instance, temperature programmed desorption (TPD) of NH₃ or CO₂ can provide an indication of the acidic or basic character of the solid catalysts. The use of these and other characterization methods such as nuclear magnetic resonance (NMR), X-ray absorption fine structure, and in situ Raman spectroscopy have already been well documented.^{23–25}

3 APPLICATIONS OF HETEROGENEOUS METAL CATALYSIS

Metal catalysis is widely used in processes related to energy production, chemicals manufacture, materials synthesis, and environmental controls, among others. In this section, both

traditional and recent examples highlighting the role of metal catalysts in each of these areas are provided.

3.1 Metal Catalysis Related to Energy Production

Metal catalysis plays a key role in the utilization of fossil fuels such as petroleum, coal, and natural gas. Petroleum technology has been identified as one of the 20 most world-shaping engineering achievements of the 20th century, supplying 40% of the global energy and 90% of the industrial organic chemicals.^{26,27} However, crude oil consists of a vast mixture of hydrocarbons (together with small amounts of sulfur and nitrogen-containing organic compounds) and therefore cannot be used efficiently without extensive processing. Components with different boiling point ranges are first separated via fractional distillation, and the naphtha fractions are then treated with hydrodesulfurization (HDS) and hydrodenitrogenation (HDN) catalysts (*see Hydrodesulfurization & Hydrodenitrogenation*) to remove impurities, the precursors of environmental pollutants such as SO₂ and NO_x, and with reforming, cracking, and hydrocracking catalysts to produce high quality fuels.

Metals, metal oxides, zeolites, and metal sulfide catalysts are at the heart of these oil conversion processes. HDS and HDN are commonly carried out on molybdenum sulfide catalysts. Catalytic cracking uses zeolites to breakdown large hydrocarbon molecules into branched or cyclic hydrocarbons and alkenes. More to the point of this review, catalytic reforming uses noble metals to promote isomerization and cyclization reactions without significantly changing the molecular weight of the naphtha fraction.^{4,28} Reforming catalysts are almost exclusively based on a platinum metallic phase. Indeed, the most common processes used nowadays are the UOP Platforming, based on Pt/Al₂O₃-SiO₂, and the Chevron Rheniforming, which employs Re-Pt/Al₂O₃-SiO₂. Typical reforming processes are carried out at temperatures between 480 and 540 °C, and with hydrogen-rich reaction mixtures under several atmospheres of pressure. The reforming catalyst is believed to rely on its dual functionality, with hydrogenation-dehydrogenation steps taking place on the metallic phase and isomerization reactions on acid sites furnished by the support, although it has been shown that platinum alone can perform all reforming steps.²⁹ Platforming and Rheniforming processes are currently used in more than 170 plants worldwide.²⁷

Coal and natural gas are also major sources of energy. Coal supplies 27% of the energy used worldwide, and natural gas, which is mainly composed of methane, provides another 23%.²⁷ At present, both coal and natural gas are primarily burned to produce energy. However, in order to fully exploit their energy and chemical potential, they are also increasingly being converted to synthesis gas, a mixture of CO and H₂, via gasification ($C + H_2O \rightarrow CO + H_2$), steam reforming ($CH_4 + H_2O \rightarrow CO + 3H_2$), or direct oxidation ($CH_4 + 1/2O_2 \rightarrow CO + 2H_2$), and further transformed into

heavier hydrocarbons and into alcohols via Fischer–Tropsch (F-T) type processes.

Again, most of the conversions enumerated above require the use of heterogeneous catalysts. F-T synthesis, for instance, uses Fe-, Co-, or Ru-based catalysts dispersed onto high surface area Al_2O_3 , SiO_2 , or TiO_2 supports. This process was initially commercialized in Germany to produce transportation fuels during World War II, and later employed in South Africa to cover their domestic fuel demands during apartheid. More recently, new F-T plants have been built in New Zealand, Malaysia, South Africa, and the Netherlands.^{30,31} The classical F-T process follows an overall reaction stoichiometry of $2n\text{CO} + 2n\text{H}_2 \rightarrow (-\text{CH}_2-)_n + n\text{CO}_2$, and typically yields a mixture of hydrocarbons with a range of hydrocarbon chain lengths. The most accepted mechanism for those reactions requires three main types of steps, namely, the initial formation of C_1 intermediates by CO dissociation and hydrogenation, the growth of hydrocarbon chains by successive insertion of C_1 building blocks into the growing hydrocarbon surface chains, and the termination and desorption of surface species.^{30,32}

Small amounts of oxygenated products such as alcohols may also be made in F-T processes depending on the catalyst used, and various parameters, including catalyst composition and operation conditions, can be used to influence the distribution of the various paraffins, olefins, and oxygenates produced. For instance, iron-based catalysts, which can be operated at high temperatures and are relatively tolerant to impurities such as sulfur in the feedstock, tend to produce straight hydrocarbon chains. Cobalt-based catalysts are even more selective, producing a much higher ratio of paraffins

to olefins and much less oxygenated products. Rhodium-containing catalysts, on the other hand, yield high fractions of alcohols, allegedly because of the slow dissociation of CO on that metal. Nowadays, F-T based gas-to-liquids (GTL) processes are used to produce diesel fuels for urban local transportation. It has been estimated that this process will supply 10% of the global diesel demand within the next 15 years.³¹

Metal catalysts are not only useful for producing gasoline, oil, diesel, and jet fuels, but also promise to contribute to the next generation of environmentally friendly energy production technology. For instance, the room temperature decomposition of liquid hydrazine (N_2H_4) to N_2 , H_2 , and NH_3 over a commercial $\text{Ir}/\gamma\text{-Al}_2\text{O}_3$ catalyst is already used as a propellant to control and adjust the orbit and attitude of satellites.³³ Perhaps more promising is the use of H_2 -fuelled electrochemical energy converters (fuel cells) in cars, space shuttles, and other power systems. This application has already been demonstrated on a commercial scale, and is predicted to reach a market in excess of 23 billion dollars by 2010.³⁴ The reaction involved, $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, is quite simple and thermodynamically favorable, but kinetically nonviable for practical applications without the use of catalysts. Figure 5 shows a typical fuel cell setup indicating the Pt-coated anode used to catalyze the oxidation of H_2 ($2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$) and the also Pt-based cathode that catalyzes the combination of the resulting protons with O_2 from air ($4\text{H}^+ + 4\text{e}^- + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$).³⁵ The two electrodes are separated by an ion-conductive electrolyte to allow for the proton transfer. The overall combination of H_2 and O_2

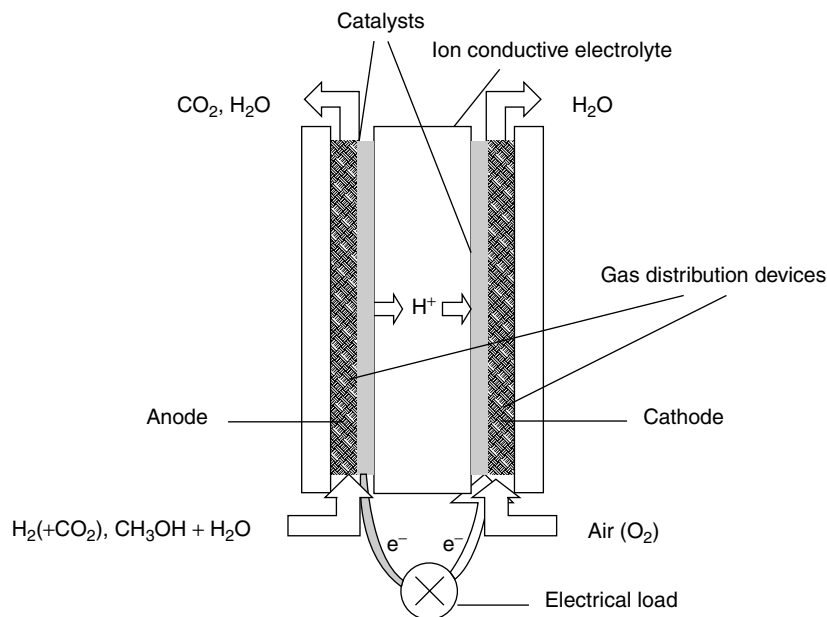


Figure 5 Schematics of a typical proton-exchange membrane fuel cell, indicating the platinum-based electrodes used for hydrogen oxidation and oxygen reduction, the interconnecting electrolyte, and the electrical circuit used to harvest the energy produced. (Reprinted from Ref. 35, © 1998, with permission from Elsevier)

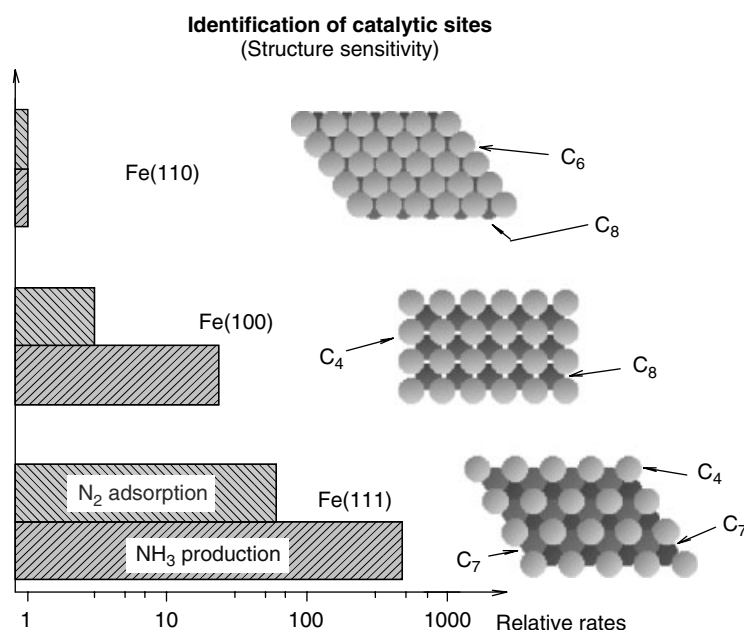


Figure 6 Correlation between both N₂ adsorption and NH₃ production rates and the structure of the catalytic surface, as determined by studies with various single crystals of iron.³⁸ These results explain the strong dependence of the performance of commercial ammonia synthesis catalysts on their method of preparation³⁹

is fast, generating an electrochemical current and additional heat. Some technical problems related to surface poisoning by CO and to fuel storage have so far limited the widespread use of fuel cell technology, but solutions to these roadblocks are already underway.³⁶ It is clear that metal catalysts will continue to play a crucial role in most future energy generation technologies.

3.2 Metal Catalysis Related to the Manufacturing of Chemicals

Metal catalysts play a pivotal role in the production of most commodity, fine, and specialty chemicals. In fact, a large number of hydrogenation and oxidation processes would not be viable without the use of heterogeneous catalysis. Perhaps the most important and mature example of a catalytic process for the production of chemicals is the synthesis of ammonia from nitrogen and hydrogen, a process that was initially commercialized in 1913 and is nowadays responsible for the production of more than 100 million tons of synthetic ammonia and associated chemicals such as nitric acid, fertilizers, explosives, caprolactam, acrylonitrile, and hydrogen cyanide each year.^{27,37} A typical ammonia synthesis catalyst is made by combining Fe₃O₄, Al₂O₃, K₂O, CaO, and MgO, and reducing this mixture in the reactor before operation to expose metallic iron sites. Surface science studies have indicated that this reaction is particularly sensitive to the structure of the active metal phase, with the dissociative adsorption of N₂, the rate-limiting step, being a couple

of orders of magnitude faster on Fe(111) planes than on Fe(100) or Fe(110) surfaces (Figure 6).^{38,39} The efficiency of the ammonia-producing process is limited by thermodynamic considerations, and reaches yields of only about 14% under the optimized reaction conditions of temperature (around 450 °C) and pressures (~100 atm). Recirculation of the reaction mixture, however, allows for ammonia yields of up to 98% under industrial settings. Supported ruthenium catalysts promoted by alkali metals have been introduced since the early 1970s to synthesize ammonia at lower temperature and pressures, and have resulted in the commercialization of the ruthenium-based Kellogg Advanced Ammonia process in Canada in 1992. Nevertheless, because of the high cost of Ru, most of the industrial ammonia plants still rely on iron catalysts.^{11,17}

Successful examples of selective oxidation catalysis in industry include the conversions of ethylene to ethylene oxide and of methanol to formaldehyde, both on silver catalysts. Ethylene oxide, with an annual worldwide production capacity over 11 million tons, is an important intermediate for the production of glycols (antifreeze agents), ethoxylates (additives in washing powder), cosmetics, polyester fibers, and pharmaceuticals.²⁷ The partial oxidation of ethylene to ethylene oxide is carried out on silver metal particles supported on α -Al₂O₃ or SiC and promoted by alkaline earth or alkali metals. Trace amounts of ethylene dichloride are also fed continuously into the reactor to suppress deep oxidation. Selectivities of about 75–85% are typical nowadays for this process. Formaldehyde, with a production capacity of

30 million tons per year, is employed for the production of the resins used as adhesives in various boards and other miscellaneous products.²⁷ This chemical is commercially produced by oxidation of methanol on electrolytic silver while feeding water vapor, nitrogen, and/or trace amounts of additives to promote methanol adsorption and inhibit deep oxidation. In addition, silver catalysts supported on low surface area pumice, ceramics, sol-gel derived SiO_2 , $\text{SiO}_2\text{-TiO}_2$, and $\text{SiO}_2\text{-Al}_2\text{O}_3$ have been used to inhibit sintering. At present the formaldehyde yield in industrial plants is 86–90%, the main by-products being carbon oxides, formic acid, and methyl formate.²⁷

The key to the success of the oxidation examples cited above is the ability of the catalysts used to exert proper kinetic control on the possible side reactions. Without it, thermodynamically favorable but undesired products such as CO_2 and H_2O are made instead. Controlling oxidation kinetics to stop at the desired oxygenated products is quite difficult, and has yet to be solved for many other systems. For instance, although many attempts have been made to develop a commercial process for the oxidation of propylene to propylene oxide, both the activity and the selectivity of the systems proposed to date, mostly based on silver catalysts, are still too low to be of industrial interest;⁴⁰ propylene oxide is presently manufactured by processes based on chlorohydrin or hydrogen peroxide instead.²⁷ In spite of these difficulties, though, recent advances in selective liquid phase oxidation of fine chemicals on supported metal catalysts have shown some promise, offering high yields (close to 100%) under mild reaction conditions.⁴¹

Other commercial approaches to the production of chemicals use selective hydrogenation of unsaturated bonds, dehydrogenation, or coupling reactions promoted by various metal catalysts.⁴² One old example related to food production is that of the manufacture of margarine by hydrogenation of unsaturated vegetable oils on nickel catalysts.^{4,17} Table 2 provides a list of other key catalytic conversion processes of commercial value in the chemical industry.^{42–46} These reactions are particularly useful for the production of ‘building blocks’ for the manufacturing of pharmaceuticals, agrochemicals, and other fine and specialty chemicals of high added value. For instance, aromatic haloamines, important intermediates in the synthesis of agrochemicals, drugs, dyestuffs, and pigments, can be manufactured via hydrogenation of aromatic halonitro compounds on supported Pt catalysts.⁴³ Ethyl (R)-2-hydroxy-4-phenylbutanoate, an intermediate for the production of benazepril, a drug used to treat high blood pressure, can be made by chiral hydrogenation of the corresponding α -ketoester on $\text{Pt/Al}_2\text{O}_3$ modified by chiral cinchona alkaloids.⁴⁴ Iminostilbene, a key intermediate in producing Carbamazepine, a medicine to treat psychiatric disorders, is made by dehydrogenation of iminodibenzyl on Pd/C catalysts.⁴³ Octyl *p*-methoxy cinnamate, the most common UV-B sunscreen in the market, can be produced via Heck coupling also using Pd/C catalysts.⁴⁵ Metal catalysts

play a central role in these and many other processes, providing highly selective pathways toward the making of desired products by activating specific functional groups in the presence of other potentially reactive moieties.

3.3 Metal Catalysis Related to the Synthesis of Materials

Heterogeneous catalysis is also extensively used for the manufacture of materials such as polymers, plastics, fibers, and carbon filaments. For example, Ziegler-Natta catalysts, based on TiCl_4 , $\text{Al}(\text{C}_2\text{H}_5)_3$ and MgCl_2 , are commercially used to fabricate many polymers, including polyethylene, polypropylene, polybutadiene, and polystyrene.¹³ Metallocene complexes have also been developed to control molecular weight and tacticity in polymerization processes. Most polymerization catalysts are homogeneous in nature (*see Oligomerization & Polymerization by Homogeneous Catalysis*), but are sometimes anchored on solid supports to facilitate the catalytic process. A more direct example of the use of heterogeneous metal catalysts in this area is that of the commercial production of caprolactam, the precursor to nylon-6 and nylon-66. This is done by using the Allied-Signal process, which involves the liquid phase hydrogenation of phenol to cyclohexanone on a palladium catalyst. That product is then converted to cyclohexanone oxime, and further to caprolactam via Beckmann rearrangement.²⁷ An alternative route for this synthesis is the Snia Viscosa process, which involves the hydrogenation of benzoic acid to hexahydrobenzoic acid on a Pd/C catalyst followed by direct conversion to caprolactam via nitrosation with nitrosylsulfuric acid.²⁷ Another practical example of the synthesis of precursors for the manufacture of materials is the hydrosilylation process, in which silicon hydrides undergoes an addition reaction with an unsaturated reactant. This reaction is particularly useful for the catalytic production, curing, and functionalization of polysiloxanes and other organosilicon polymers. Although homogeneous metal complexes are often used for these hydrosilylation reactions, supported Pt catalysts are preferred in many applications.⁴⁶

Carbonaceous deposits have been long regarded as unwanted side products during catalysis.⁴⁷ However, with the advent of nanotechnology, there has been a renewed interest in the use of metal catalysts for the production of filamentous carbon materials such as carbon nanotubes (*see Carbon: Nanotubes*) and nanofibers because of their potential applications in electron emission devices, hydrogen-storage materials, fuel cell electrodes, and catalyst supports.⁴⁸ Physical synthetic methods such as arc-discharge and laser ablation are already available for the production of carbon filaments, but their yields are relative low. Metals such as Fe, Co, and Ni, on the other hand, can be used to convert CH_4 , CO, synthesis gas, C_2H_2 , C_2H_4 , or C_2H_6 efficiently to carbon nanofilaments or composite materials in good quantities. Figure 7 provides

Table 2 Important hydrogenation, dehydrogenation, and coupling processes catalyzed by metals⁴²⁻⁴⁶

Process	Example reaction scheme	Metal catalyst
Hydrogenation of alkenes and alkynes		Pd, Rh, Pt
Hydrogenation of aromatic rings		Rh, Ru, Pt
Hydrogenation of aldehydes, acetones, and carboxylic acids to alcohols		Pt, Ru, Pd, Ir
Hydrogenation of nitro and nitroso compounds to amines		Pd, Pt
Hydrogenation of nitriles to amines		Rh, Pt, Pd
Hydrogenation of oximes to amines, hydroxyamines or imines		Rh, Pt, Pd
Hydrogenation of imines to amines		Pt
Enantioselective hydrogenation of α - or β -ketoesters		Cinchona-modified Pt, Pd, or tartaric acid-modified Ni

(cont'd overleaf)

Table 2 cont'd

Process	Example reaction scheme	Metal catalyst
Hydrodehalogenation		Pd
Dehydrogenation		Pd, Pt, Rh
Hydrogenolysis of sulfides		Re
Reductive alkylation		Pt, Pd
Arylation or vinylation of alkenes with aryl or vinyl halides (Heck coupling)		Pd
Suzuki coupling of arylboronic acids and aryl halides to biaryls		Pd
Hydrosilylation		Pt

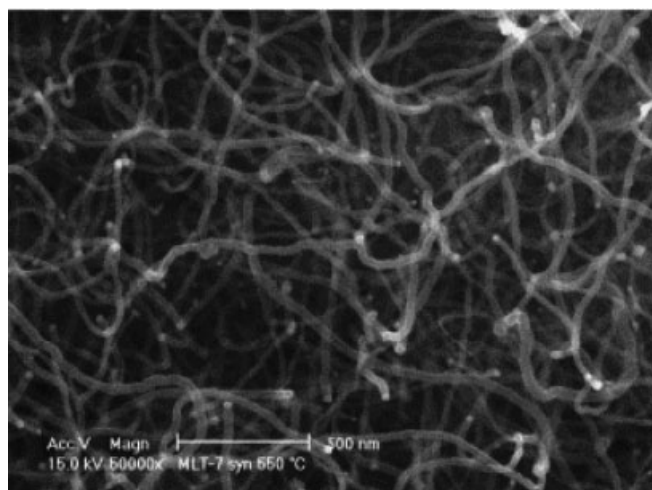


Figure 7 SEM image of carbon nanofibers fabricated using a Ni/SiO₂ catalyst. The nickel catalytic particles are seen here as capping light-colored spheres. Carbon nanotubes grow by decomposition of hydrocarbons on the surface of those metal particles followed by insertion into the carbon growing structures. (Reprinted from Ref. 49. © 2003, with permission from Elsevier)

an example of the nanofibers that can be grown this way.⁴⁹ It shows both the grown carbon fibers and the capping metal particles that catalyze the carbon deposition process. The rate-determining step in most of those catalytic reactions is thought to be the diffusion of carbon atoms through metal particles, and the properties of the resultant products are believed to depend on the composition and nature of the catalyst as well as on operational parameters such as temperature, reaction time, composition of reactant mixture, and gas flow rate.⁴ Other carbon materials such as diamond can also be synthesized with the aid of metal alloy catalysts.⁵⁰ Unfortunately, full understanding of these processes is at present limited, and the production and practical use of well-defined carbon fibers and diamond materials by catalytic means are still in the future.

3.4 Metal Catalysis Related to Environmental Control

Metal catalysis is not only good for the production of chemicals, but has also been increasingly employed for the control of pollutant emissions and for the removal of contaminants. In fact, a third of the 9–10 billion dollars market for world catalyst consumption nowadays is related to environmental catalysis.^{2,51} One of the most prominent uses of environmental catalysis is in the control of automobile exhausts. Stringent regulations on car gas emissions have now been imposed by numerous governments around the world to minimize the buildup of noxious CO, NO_x, and hydrocarbon gases in the troposphere. To meet these standards, modern internal combustion engines are fitted with commercial catalytic converters designed to transform those pollutants

to less toxic N₂, H₂O, and CO₂. The typical so-called three-way catalyst used for this in most countries is composed of a combination of Pt, Rh, and Pd metal particles supported on a γ -Al₂O₃ washcoat stabilized with La₂O₃ or BaO and modified with CeO₂ or CeO₂/ZrO₂. These catalysts are set onto a ceramic honeycomb or monolith to obtain high thermal stability, high heat and mass transfer rates, low pressure drops, and long catalyst life (Figure 8).⁵²

The conversion of automobile exhaust gases involves a number of complex oxidation and reduction reactions, which with today's catalysts requires careful control of the air/fuel ratio within a narrow window around the stoichiometric value of 14.6. Direct injection gasoline engines, however, operate best with a large excess of air, and that compromises the reduction efficiency for NO_x. In order to address this problem, alkaline metal oxides such as BaO are being incorporated into new catalytic designs to adsorb NO_x; periodic brief incursions into rich fuel-burn operation conditions are then induced to release and reduce the trapped NO_x on the metal catalyst.⁵³ Another technical problem in the present systems is their ineffectiveness during the so-called cold start, right after the engine is turned on; this is the time when most of the polluting hydrocarbon emissions takes place. The challenge here is to either develop ways to achieve fast heating of the catalyst to its light-off temperature or to design catalysts more active under lower operating temperatures.⁵³ New metal-based formulations have already been incorporated in the new generations of automobile catalytic converters.

Three-way catalysts are mainly used to treat gaseous pollutants from mobile engines. In stationary sources such as power generators and industrial plants for the production of nitric acid and other chemicals, the most important requirements in terms of environmental control are the removal of NO_x and volatile organic compounds (VOCs) from gas discharges. In general, NO_x is selectively reduced (SCR) on either V₂O₅-WO₃/TiO₂ or metal-exchanged zeolite catalysts with NH₃ or urea. However, the use of nitrogen-containing reducing agents can lead to the production of other pollutants, so recent attempts have been made to use metal catalysts such as Pd/TiO₂ together with methane as the reducing agent instead.⁵⁴ In terms of the treatment of VOCs, they are often burned over Pt/Al₂O₃-BaO or Pd/Al₂O₃ catalysts in order to lower the temperature of the process well below those used in conventional flame combustion (up to 2000 °C).⁵¹

Customized catalysts are also available for more specialized environmental uses. For instance, Pt/mordenite catalysts are used for the treatment of air pollutants in spray painting and coating workshops, and also to control emission in benzoic anhydride manufacturing industries. Pt-V₂O₅-SO₄²⁻/mordenite is sometimes employed to purify flue gases containing sulfur compounds. Metal-based catalysts are ideal for such applications, because they work at low reactant concentrations (4–6 g/m³), high space velocities

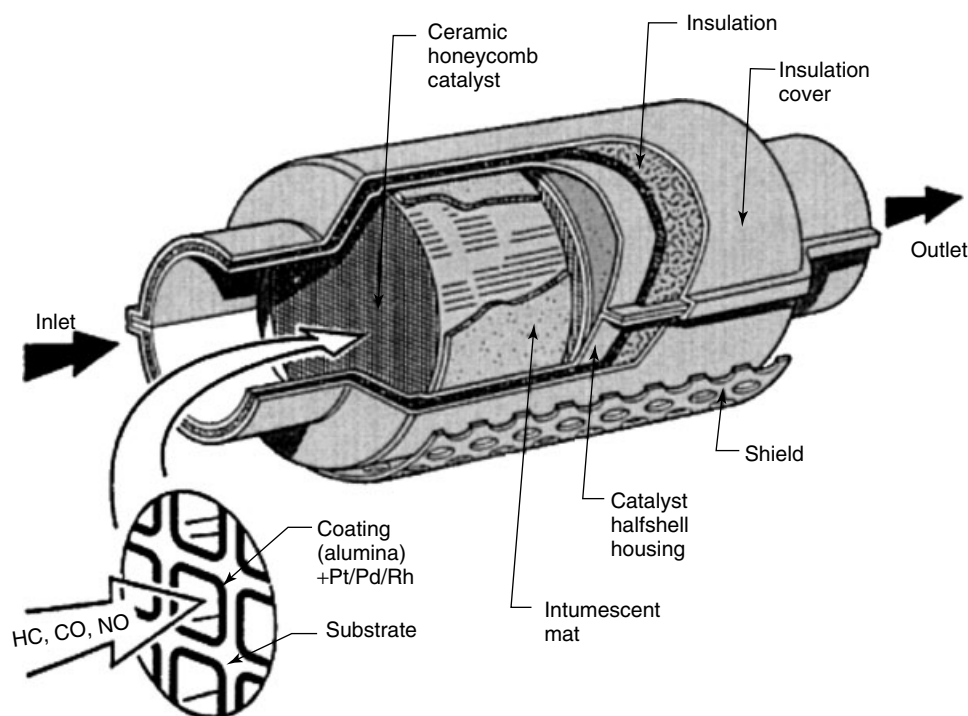


Figure 8 Typical design of a three-way catalyst for automobile exhaust control. Highlighted here are the honeycomb support and the mounting can used. So-called three-way catalysts, consisting of a combination of Pt, Rh, and Pd particles dispersed on high surface area alumina, are spread on the honeycomb structure to oxidize the carbon monoxide and unburned hydrocarbons and to reduce the nitrogen oxides released by the engine of the car. (Reprinted from Ref. 52, © 1998, with permission from Elsevier)

(4500–20 000 h⁻¹), and low temperatures (150–400 °C). Processes such as those mentioned above have been adopted in more than 1000 factories in China and Japan alone.⁵⁵

Another use of environmental catalysis is in the hydrodechlorination of chlorine-containing pollutants such as chlorinated olefins, chlorobenzenes, chlorophenols, and chlorofluorocarbons. Thermal combustion of these chemicals has been demonstrated on a commercial scale, but the required high temperatures for incineration lead to the formation of toxic dioxins. Alternatively, supported Pd, Pt, Rh, Ni, Pd–Au and Pd–Re catalysts can be used to remove the chlorine atoms from those molecules via hydrogenation reactions.^{56,57} Hydrodechlorination is also one of the purifying steps needed to improve the quality of drinking water and to clean waste water.⁵⁸ In another example, zero-valent iron has been used as an electrocatalyst for the removal of both organic chlorinates and inorganic salts such as arsenates from groundwater.⁵⁹ Wet air oxidation is used to remove dissolved organic pollutants, and hydrodenitrification is used to abate inorganic salts such as nitrates and nitrites in water.⁵⁸ Other applications of metal catalysis in this area include the indoor detection and room temperature oxidation of carbon monoxide, the elimination of odors in indoor environments, and the decomposition of aqueous ozone.⁵¹

4 PROMISING NEW DIRECTIONS IN HETEROGENEOUS METAL CATALYSIS

The field of catalysis is a mature one, with a history over 100 years old. Nevertheless, new opportunities keep on developing in this area. Below, some of the new directions that we believe will lead the near future of heterogeneous catalysis are advanced.

4.1 Novel Catalytic Materials

One of the main problems of traditional metal catalysts is that they are fairly ill-defined, in that the shapes of the metal particles on the supports are often irregular, that their particle size distributions are broad, and that no strict control can be exerted on interparticle distances. Recent developments in material technologies promise to provide new metallic materials, new supports, and new methodologies for the fabrication of better defined catalysts.

One family of new materials with potential use in metal catalysis is that of metal nanostructures such as metal nanoparticles (see *Metal Nanoparticles, Synthesis of* and *Metal Nanoparticles, Organization & Applications of*), nanoshells, nanowires, nanorods, nanotubes, nanobelts, and nanoplates.^{60–62} For instance, it has been recently shown that Ag nanowires and nanoparticles can be produced by

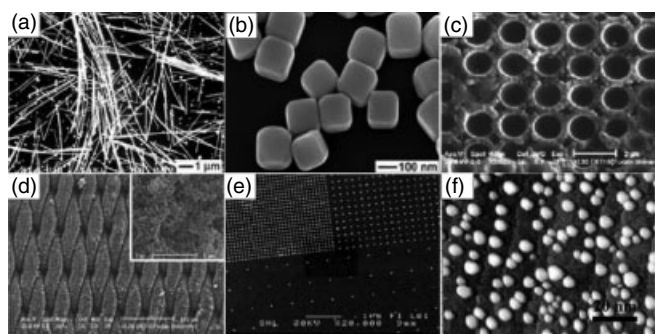


Figure 9 Examples of novel materials with potential catalytic applications. From left to right and top to bottom, these pictures represent: (a) Ag nanowires. (Reprinted with permission from Ref. 63. © 2002 American Chemical Society); (b) Ag nanoparticles. (Reprinted with permission from Y. Sun and Y. Xia, *Science*, 2002, **298**, 2176. © 2002 AAAS (www.sciencemag.org)); (c) zeolite monolith. (Ref. 67. Reproduced by permission of Kluwer Academic/Plenum Publishers); (d) zeolite coatings on stainless steel grids. (Ref. 68. Reproduced by permission of Wiley-VCH); (e) arrays of Pt nano lithography-made particles on SiO_2 . (Ref. 70. Reproduced by permission of Kluwer Academic/Plenum Publishers); and (f) Ag nanoparticles vapor deposited on an Al_2O_3 thin film⁷¹

nucleation and growth on nanoseeds produced by first reducing PtCl_2 or AgNO_3 with ethylene glycol (Figures 9a and 9b).^{63,64} Moreover, those Ag nanowires can be further used as templates to grow Au, Pt, or Pd nanotubes with hollow interiors.⁶² The controlled growth of other metals with various nanostructures have been reported in the recent literature, but their uses in catalysis are still in the embryonic stages. One tantalizing result in this direction is that of the use of palladium nanoshells and nanotubes to catalyze the Suzuki coupling reaction with up to 100% yield.^{61,62}

Because of their ordered pore structures and adjustable acid-base properties, zeolites have long been used as supports for many catalysts. However, the small size of the micropores in zeolites (diameter < 1 nm) limits their application when dealing with bulky molecules.^{16,65} The recent development of mesoporous metal oxides (*see Porous Inorganic Materials*) with ordered nano channels promises to allow for the engineering of 'nano reactors' for reactions involving fine chemicals and pharmaceuticals. Organometallic compounds, metal oxides, heteropolyacids, and enzymes have all been supported onto mesoporous silica, but the application of mesoporous supports in metal catalysis has only started.^{65,66} Advanced zeolite-based architectures such as zeolite shells, fibers, tissues, monoliths, capsules, membranes, and coatings as well as mesoporous materials with sturdy zeolite walls have also been constructed by assembly of nanozeolite building blocks (Figures 9c and 9d).^{67–69} These new materials with ordered secondary advanced structures are expected to allow for fast mass transfer during catalysis.

Physical methods such as those developed for the semiconductor industry have also started to make an inroad into metal catalysis.^{38,70,71} Figures 9e and 9f show micrographs of metal particles supported on metal oxide thin films fabricated by electron beam lithography and by metal vapor deposition, respectively.^{70,71} As exemplified by Figure 9e, the size, height, and spatial distribution of the metal particles may be easily adjusted with these techniques according to experimental needs. Although applications of these new materials technologies to the manufacturing of new commercial catalysts are still somewhere in the future, they already offer new opportunities for the study of reaction kinetics on supported metal particles with unique and uniform structures.

4.2 Surface Science of Catalysis

The development of new catalytic materials needs to be complemented with detailed studies of the surface chemistry of catalysis at the molecular level in order to better define the requirements for the catalytic active sites. The wide array of modern spectroscopies available to surface scientists today is ideally suited for this task (*see Surfaces*). Surface science studies on catalysis typically probe reaction intermediates on model metal samples under well controlled conditions. This kind of study is traditionally carried out in ultrahigh vacuum (UHV) systems such as that shown in Figure 10. Single crystals or other well-defined metal surfaces are cleaned and characterized in situ by physical and chemical means, and then probed using a battery of surface sensitive techniques such as photoelectron (XPS and UPS), electron energy loss (ELS

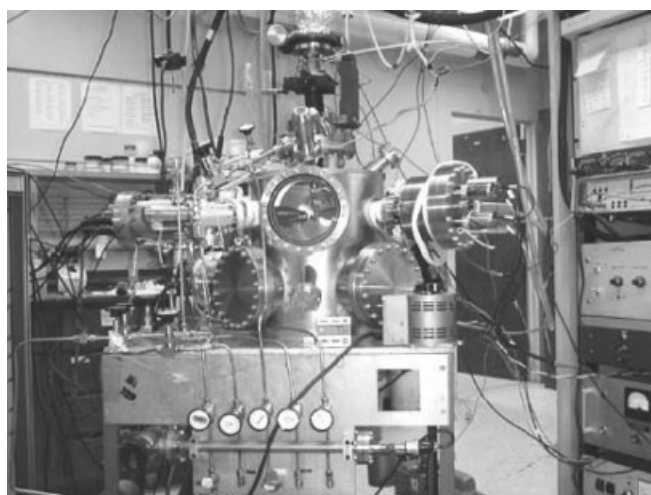


Figure 10 Typical ultrahigh vacuum system used for surface science studies of catalytic reactions on model systems.⁷⁴ The combined development of new preparation methods for realistic catalytic samples and in situ spectroscopies for the molecular level characterization of surface species during catalysis promises to advance the basic understanding of catalytic processes

and HREELS), secondary ion mass (SIMS), infrared (IR), and temperature programmed desorption (TPD) spectroscopies. By characterizing the catalytic surface before, during, and after chemical reactions using a combination of these surface science methods, a molecular picture of the reaction pathways of catalytic processes on solid surfaces can be extracted.^{38,72–74}

Although traditional surface science studies have greatly helped advance our understanding of catalysis at a fundamental level, the differences between the pressure and materials employed in UHV studies and those encountered in applied catalysis have limited the extrapolation of the knowledge developed this way to real systems. Ex-situ characterization experiments can be carried out on catalytic surfaces before and after reactions, but those are also subject to limitations due to potential changes during the handling of the sample. This is why there has been a renewed interest in closing these so-called ‘materials’ and ‘pressure’ gaps between surface science and catalysis by using new spectroscopic and material preparation methods. To bridge the materials gap, novel model catalysts consisting of metal particles supported on planar metal oxide thin films have started to be developed and tested for a number of simple catalytic reactions such as ethylene hydrogenation, CO methanation, CO oxidation, ethane hydrogenolysis, NO reduction, and methanol decomposition.^{38,75,76} To bridge the pressure gap, in situ spectroscopic methods such as sum frequency generation (SFG), Fourier transform infrared (FT-IR), and Raman spectroscopies as well as scanning (STM) and electron (SEM, TEM) microscopies adopted for *in situ* operation under nonvacuum conditions have been employed to study reactions on metal surfaces under realistic conditions.⁷⁷ These approaches promise to advance our fundamental

chemical understanding of complex catalytic systems at the molecular level.

4.3 High Throughput Catalyst Testing

Because of the complexity of catalytic systems and the lack of understanding of their microscopic properties, the design of catalysts from first principles is still limited. Instead, the traditional way of developing successful catalysts in industry has been by trial and error. This approach is not only costly, but also time-consuming. Consequently, to expedite the development of new catalysts, intensive efforts have been recently placed on automating their production and characterization during the first stages of screening. With these high throughput (sometimes called combinatorial) techniques, small amounts of many catalysts with slightly different and systematically varied compositions are prepared in parallel, and simultaneously subjected to reaction testing by techniques such as scanning mass spectrometry, resonance-enhanced multiphoton ionization, fluorescence detection, and IR thermography. The data collected are subsequently analyzed with the aid of computers and sophisticated software to mine promising leads from the thousands of candidates. The best cases are then scaled up, optimized, and subjected to a more quantitative second screening to select even better leads for further scaling up and testing in conventional lab reactors.⁷⁸

High throughput catalyst screening is especially useful for the development of metal catalysts with multiple components such as alloys. Figure 11 shows the results of a secondary screening of four-component catalysts for the

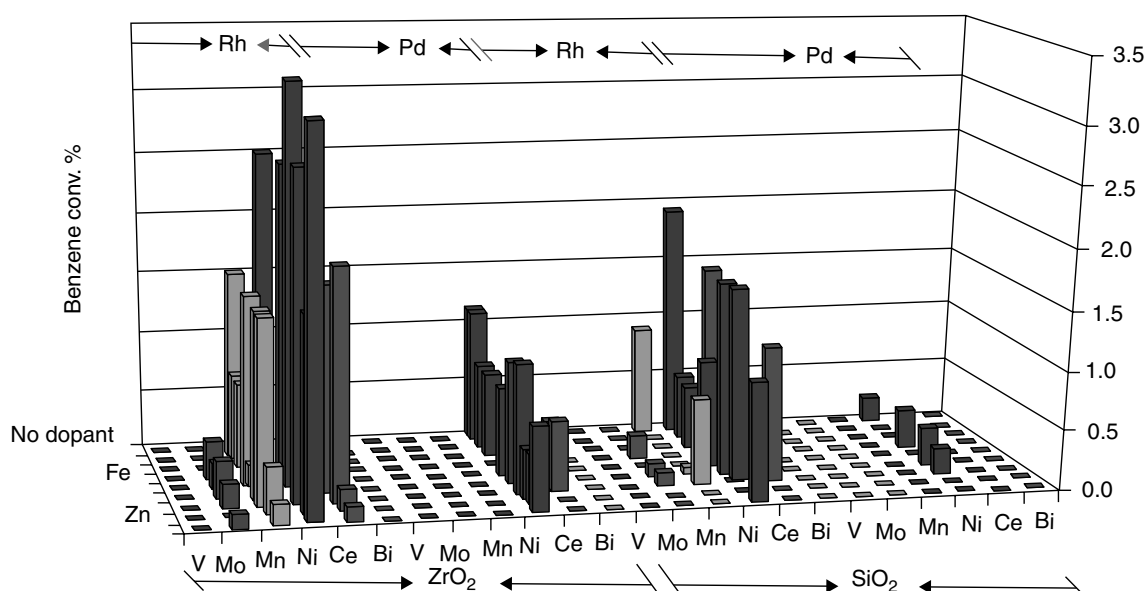


Figure 11 Results from high throughput studies on four-component catalysts for the direct amination of benzene to aniline. A library with (Rh, Pd)– M_1 – M_2 –(ZrO_2 , SiO_2) components was tested. New technology for the fast screening of complex materials offers an opportunity to test a wide variety of potential catalysts for specific catalytic applications. (Reprinted from Ref. 79, © 1998, with permission from Elsevier)

direct amination of benzene to aniline. The library used there consists of combinations of an active metal (Rh or Pd), a support (ZrO_2 or SiO_2), an oxidant (V, Mo, Mn, Ni, Ce, or Bi), and a dopant (blank, V, Cr, Mn, Fe, Co, Ni, Cu, or Zn).⁷⁹ 216 samples generated by the combination of these four components were tested 24 samples at a time by using parallel high-pressure batch reactors. It became clear from these data that ZrO_2 is a better support than SiO_2 , that Rh is a better metal than Pd, that NiO is the best oxidant, and that metal dopants can further modify the overall activity of the system.⁷⁹ The use of combinatorial methods for the development of a few other catalysts for processes such as the oxidation of CO by O_2 or NO with Rh–Pd–Pt alloys has been reported as well. Nevertheless, their usefulness for the design of commercial catalysts is still in question.^{78,80}

4.4 Improved Selectivities, Green Chemistry

Historically, the emphasis in catalysis has been on increasing activity rather than selectivity. This is not surprising, since catalysts are generally regarded as substances that can speed up reactions. In addition, many early catalytic processes were based on relatively simple reactions with negligible sidesteps. As the field of catalysis expanded and covered more complex reactions, however, selectivity started to become crucial in designing industrial processes.⁸¹ One example of this already mentioned above is that of the partial oxidation of alcohols to oxygenated hydrocarbons.⁸²

A more subtle but important example of the need for good catalytic selectivity is that of the manufacture of chiral medicines. The market for single-enantiomer drugs has recently reached 160 billion dollars, accounting for close to half of the profits from sales of drugs worldwide.⁸³ If regular catalysts are used, only racemic mixtures of the products are obtained. This is at the very least quite wasteful, since those processes require at least twice the amount of reactants. Moreover, in many instances, one enantiomer may be biologically active but the other may be poisonous, so the desired product needs to be isolated by expensive separation techniques such as selective crystallization, chiral chromatography, membrane separation, or chemical methods. Clearly, the production of enantiomerically pure chemicals directly via heterogeneous catalysis would be much preferred. In this direction, tartaric acid-modified nickel and cinchona-modified platinum catalytic systems have proven successful for the enantioselective hydrogenation of β - and α -ketoesters, respectively, showing enantioselectivities of up to 98%.^{44,84} The role of the chiral modifiers is still not fully understood, but in the latter case appears to involve the formation of a 1:1 complex with the reactant on the surface to force an adsorption geometry where the hydrogenation is only possible on one side of the molecule (Figure 12).⁸⁴ The use of heterogeneous catalysis in chiral

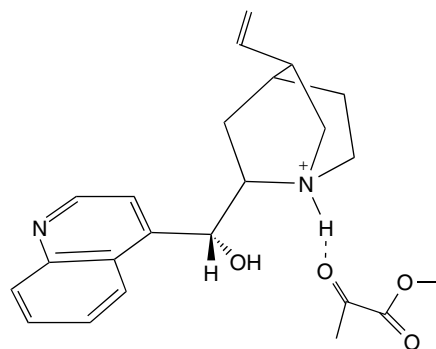
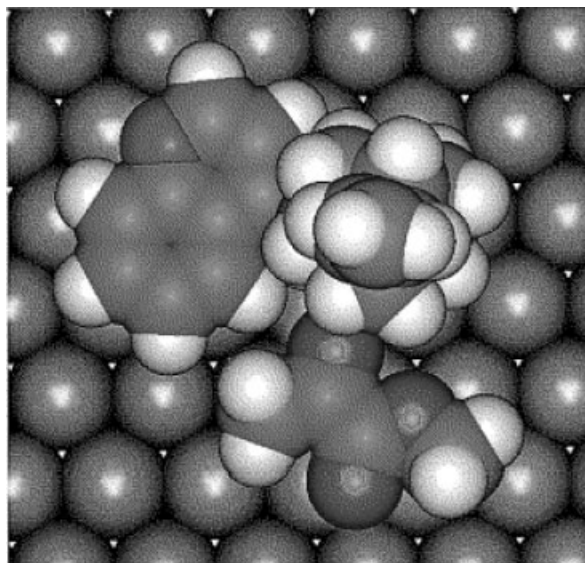


Figure 12 Schematic representation of the mechanism proposed for the enantioselective hydrogenation of α -ketoesters on cinchonidine-modified platinum catalyst. The development of general methods for imparting enantioselectivity to regular heterogeneous catalysts promises to revolutionize the pharmaceutical and agrochemical industries.⁸⁴

manufacturing is still quite limited, but ongoing work to try to understand the details of this chemistry promises to lead to the development of new catalysts for these applications.⁸⁵

Many industrial catalytic processes in use nowadays are far from ideal in terms of cleanliness, economy, and simplicity. For one, multiple-steps processes often require the disposal of many side products, and display low compounded yields. With the introduction of the concept of green chemistry in the 1990s, whereby a new emphasis is placed on the design of chemical products and processes that reduce or eliminate the use and generation of hazardous substances, significant efforts are being placed to develop alternative more efficient routes to replace some existing processes in industry.^{86,87} Heterogeneous metal catalysis is expected to play a crucial role in these new processes as well.

5 REFERENCES

1. R. A. van Santen, P. W. N. M. van Leeuwen, J. A. Moulijn, and B. A. Averill eds, 'Catalysis: An Integrated Approach', Elsevier, Amsterdam, 1999.
2. North American Catalysis Society, http://www.nacatsoc.org/edu_catalysis.asp, 2005.
3. G. C. Bond, 'Catalysis by Metals', Academic Press, London, 1962.
4. C. N. Satterfield, 'Heterogeneous Catalysis in Industrial Practice', Krieger Publishing, Malabar, 1996.
5. F. Zaera, *Acc. Chem. Res.*, 2002, **35**, 129.
6. G. C. Bond, 'Heterogeneous Catalysis: Principles and Applications', Clarendon Press, Oxford, 1974.
7. J. M. Thomas and W. J. Thomas, 'Introduction to the Principles of Heterogeneous Catalysis', Academic Press, London, 1967.
8. M. Boudart and G. Djéga-Mariadassou, 'Kinetics of Heterogeneous Catalytic Reactions', Princeton University Press, Princeton, NJ, 1984.
9. W. J. M. Rootsaert and W. M. H. Sachtler, *Z. Phys. Chem. N.F.*, 1960, **26**, 16.
10. B. C. Gates, 'Catalytic Chemistry', John Wiley & Sons, New York, 1992.
11. A. Ozaki and K. Aika, in 'Catalysis: Science and Technology', eds. J. R. Anderson and M. Boudart, Springer-Verlag, New York, 1981, Vol. 1, p. 87.
12. G. A. Somorjai, 'Introduction to Surface Chemistry and Catalysis', John Wiley & Sons, New York, 1994.
13. B. Cornils, W. A. Herrmann, R. Schlögl, and C. H. Wong eds, 'Catalysis from A to Z', Wiley-VCH, Weinheim, 2000.
14. F. Zaera, *J. Phys. Chem. B*, 2002, **106**, 4043.
15. E. A. Sales, B. Benhamida, V. Caizergues, J.-P. Lagier, F. Fiévet, and F. Bozon-Verduraz, *Appl. Catal. A*, 1998, **172**, 273.
16. M. E. Davis, *Ind. Eng. Chem. Res.*, 1991, **30**, 1675.
17. G. Ertl, H. Knözinger, and J. Weitkamp eds, 'Handbook of Heterogeneous Catalysis', VCH, Weinheim, 1997.
18. C. Perego and P. Villa, *Catal. Today*, 1997, **34**, 281.
19. F. Pinna, *Catal. Today*, 1998, **41**, 129.
20. R. D. Gonzalez, T. Lopez, and R. Gomez, *Catal. Today*, 1997, **35**, 293.
21. R. B. Anderson and P. T. Dawson eds, 'Experimental Methods in Catalytic Research', Academic Press, New York, 1976, Vol. 3.
22. G. Leofanti, G. Tozzola, M. Padovan, G. Petrini, S. Bordiga, and A. Zecchina, *Catal. Today*, 1997, **34**, 307.
23. J. W. Niemantsverdriet, 'Spectroscopy in Catalysis', Wiley-VCH, Weinheim, 2000.
24. J. F. Haw ed., 'In Situ Spectroscopy in Heterogeneous Catalysis', Wiley-VCH, Weinheim, 2002.
25. F. Zaera, in 'Encyclopedia of Chemical Physics and Physical Chemistry', eds. J. Moore and N. Spencer, Institute of Physics Publishing, Bristol, 2001, Vol. 2, p. 1563.
26. US National Academy of Engineering, <http://www.greatachievements.org>, 2000.
27. Kirk-Othmer ed., 'Kirk-Othmer Encyclopedia of Chemical Technology', 4th ed., Wiley-Interscience, New York, 1992–1998, Vols. 1–27.
28. J. H. Sinfelt, in 'Catalysis: Science and Technology', eds. J. R. Anderson and M. Boudart, Springer-Verlag, New York, 1981, Vol. 1, p. 257.
29. F. Zaera and G. A. Somorjai, in 'Hydrogen Effects in Catalysis: Fundamental and Practical Applications', eds. Z. Paál and P. G. Menon, Marcel Dekker, New York, 1988.
30. M. E. Dry, *Catal. Today*, 2002, **71**, 227.
31. A. H. Tullo, *Chem. Eng. News*, 2003, **81**(29), 18.
32. P. Biloen and W. M. H. Sachtler, *Adv. Catal.*, 1981, **30**, 165.
33. S. Balcon, S. Mary, C. Kappenstein, and E. Gengembre, *Appl. Catal. A*, 2000, **196**, 179.
34. M. Jacoby, *Chem. Eng. News*, 2003, **81**(3), 32.
35. P. M. Urban, A. Funke, J. T. Müller, M. Himmen, and A. Doctor, *Appl. Catal. A*, 2001, **221**, 459.
36. L. Carrette, K. A. Friedrich, and U. Stimming, *Chem. Phys. Chem.*, 2000, **1**, 162.
37. M. Grunze, in 'The Chemistry and Physics of Solid Surface and Heterogeneous Catalysis', eds. D. A. King and D. P. Woodruff, Elsevier, Amsterdam, 1982, Vol. 4, p. 143.
38. G. A. Somorjai, *J. Phys. Chem. B*, 2000, **104**, 2969.
39. F. Zaera, *Surf. Sci.*, 2002, **500**, 947.
40. J. R. Monnier, *Appl. Catal. A*, 2001, **221**, 73.
41. M. Besson and P. Gallezot, *Catal. Today*, 2000, **57**, 127.
42. R. L. Augustine, *Catal. Today*, 1997, **37**, 419.
43. Johnson Matthey Catalysts, <http://www.chemicals.matthey.com/catalysts/introhet.asp>, 2001.
44. H. U. Blaser, F. Spindler, and M. Studer, *Appl. Catal. A*, 2001, **221**, 119.
45. H. U. Blaser, A. Indolese, A. Schnyder, H. Steiner, and M. Studer, *J. Mol. Catal. A*, 2001, **173**, 3.
46. B. Marciniak, *Silicon Chem.*, 2002, **1**, 155.
47. S. M. Davis, F. Zaera, and G. A. Somorjai, *J. Catal.*, 1982, **77**, 439.
48. K. P. de Jong and J. W. Geus, *Catal. Rev.-Sci. Eng.*, 2000, **42**, 481.
49. M. L. Toebes, F. F. Prinsloo, J. H. Bitter, A. J. van Dillen, and K. P. de Jong, *J. Catal.*, 2003, **214**, 78.
50. Y. Li, Y. Qian, H. Liao, Y. Ding, L. Yang, C. Xu, F. Li, and G. Zhou, *Science*, 1998, **281**, 246.
51. G. Centi, P. Ciambelli, S. Perathoner, and P. Russo, *Catal. Today*, 2002, **75**, 3.
52. R. M. Heck and R. J. Farrauto, *Appl. Catal. A*, 2001, **221**, 443.

53. M. Shelef and R. W. McCabe, *Catal. Today*, 2000, **62**, 35.
54. M. D. Fokema and J. Y. Ying, *Catal. Rev.*, 2001, **43**, 1.
55. Z. Gao, *Stud. Surf. Sci. Catal.*, 1998, **121**, 143.
56. M. Makkee, A. Wiersma, E. J. A. X. van de Sandt, H. van Bekkum, and J. A. Moulijn, *Catal. Today*, 2000, **55**, 125.
57. G. Yuan and M. A. Keane, *Catal. Commun.*, 2003, **4**, 195.
58. Y. I. Matatov and M. Sheintuch, *Ind. Eng. Chem. Res.*, 1998, **37**, 309.
59. J. Farrell, J. Wang, P. O'Day, and M. Conklin, *Environ. Sci. Technol.*, 2001, **35**, 2026.
60. M. A. El-Sayed, *Acc. Chem. Res.*, 2001, **34**, 257.
61. S.-W. Kim, M. Kim, W. Y. Lee, and T. Hyeon, *J. Am. Chem. Soc.*, 2002, **124**, 7642.
62. Y. Sun, B. Mayers, and Y. Xia, *Adv. Mater.*, 2003, **15**, 641.
63. Y. Sun, Y. Yin, B. T. Mayers, T. Herricks, and Y. Xia, *Chem. Mater.*, 2002, **14**, 4736.
64. Y. Sun and Y. Xia, *Science*, 2002, **298**, 2176.
65. A. Corma, *Chem. Rev.*, 1997, **97**, 2373.
66. D. T. On, D. Desplantier-Giscard, C. Danumah, and S. Kaliaguine, *Appl. Catal. A*, 2003, **253**, 545.
67. Y. Wang, Y. Tang, X. Wang, W. Shan, C. Ke, Z. Gao, J. Hu, and W. Yang, *J. Mater. Sci. Lett.*, 2001, **20**, 2091.
68. A. Dong, Y. Wang, Y. Tang, Y. Zhang, N. Ren, and Z. Gao, *Adv. Mater.*, 2002, **14**, 1506.
69. Y. Liu and T. J. Pinnavaia, *J. Mater. Chem.*, 2002, **12**, 3179.
70. B. Kasemo, S. Johansson, H. Persson, P. Thormählen, and V. P. Zhdanov, *Top. Catal.*, 2000, **13**, 43.
71. X. Lai and D. W. Goodman, *J. Mol. Catal. A*, 2000, **162**, 33.
72. D. P. Woodruff and T. A. Delchar, 'Modern Techniques of Surface Science', Cambridge University Press, Cambridge, MA, 1988.
73. C. T. Campbell, *Adv. Catal.*, 1989, **361**, 1.
74. F. Zaera, *Prog. Surf. Sci.*, 2001, **69**, 1.
75. D. W. Goodman, *J. Catal.*, 2003, **216**, 213.
76. H.-J. Freund, M. Bäumer, J. Libuda, T. Risse, G. Rupprechter, and S. Shaikhudinov, *J. Catal.*, 2003, **216**, 223.
77. G. A. Somorjai, *CATTECH*, 1999, **3**, 84.
78. B. Jandeleit, D. J. Schaefer, T. S. Powers, H. W. Turner, and W. H. Weinberg, *Angew. Chem., Int. Ed. Engl.*, 1999, **38**, 2494.
79. A. Hagemeyer, R. Borade, P. Desrosiers, S. Guan, D. M. Lowe, D. M. Poojary, H. Turner, H. Weinberg, X. Zhou, R. Armbrust, G. Fengler, and U. Notheis, *Appl. Catal. A*, 2002, **227**, 43.
80. A. H. Tullo, *Chem. Eng. News*, 2003, **81**(29), 14.
81. F. Zaera, *Catal. Lett.*, 2003, **91**, 1.
82. F. Zaera, *Catal. Today*, 2003, **81**, 149.
83. A. M. Rouhi, *Chem. Eng. News*, 2003, **81**(18), 45.
84. A. Baiker, *J. Mol. Catal. A*, 2000, **163**, 205.
85. Z. Ma, I. Lee, J. Kubota, and F. Zaera, *J. Mol. Catal. A*, 2004, **216**, 199.
86. W. F. Höelderich, *Catal. Today*, 2000, **62**, 115.
87. P. T. Anastas, M. M. Kirchhoff, and T. C. Williamson, *Appl. Catal. A*, 2001, **221**, 3.